Java Internship at Innowise Group

Java core

1. How to create a thread?
2. What is the difference between primitive and reference data types?
3. What can the final keyword be applied to?
4. What is static and why is it needed?
5. What is a functional interface? Name the popular ones
6. What is a terminal function?
7. Map vs flatMap stream API
8. OOP principles
9. Contract equals and hashcode
10. How does inheritance work in java?
11. What are generic types and why are they needed?
12. Methods of the Object class. Name them and tell why they are needed
13. The most popular GCs and general algorithm of work
14. What is try-with-resources and why is it needed?
15. Int pool and String pool. What is it and why is it necessary?
16. Stack and Heap, what is stored and where.
17. Popular thread-safe collections. Name and tell how they work
18. HashMap algorithm
19. Differences between ArrayList and LinkedList. What and when works faster.
20. How do TreeMap and TreeSet work?
21. Hierarchy of collections. Tell us what it looks like and name the implementations
22. Exception hierarchy. Checked and Unchecked exceptions
23. What are wrapper classes and why are they needed?
24. Immutable objects. Examples in Java. How to create them?
25. What is serialization in Java? What do you need to do to serialize an object? What is serialVersionUID?
26. What is the volatile keyword for?
27. Is an array an object or a primitive?
28. Thread.sleep vs Thread.yield разница
29. How do the wait, notify, notifyAll methods work?
30. How does the Thread.join() method work?
31. What is AtomicInteger and how does it work?
32. Why is the transient keyword needed?
33. What are deadlock and livelock?
34. Methods for synchronizing access to shared resources between threads. Name them and name the advantages and disadvantages
35. What is the synchronized keyword used for and what are its features?

SQL

1. Types of JOIN. Name and tell how they differ.
2. HAVING and WHERE distinctions
3. What is a primary key in SQL and why is it important?
4. What is a foreign key and how is it used to establish relationships between tables?
5. How to build a many-to-many connection?
6. What is the difference between the DELETE and TRUNCATE statements?
7. What is normalization and why is it needed? Name the number of forms of normalization
8. What is a transaction and why is it needed?
9. ACID. Name the principles and levels of transaction isolation.
10. Main types of indexes and how they work
11. What is a query execution plan and how to view it

Spring:

1. Bean lifecycle. What it looks like, how the bean is created, etc.
2. Methods for declaring beans. List and name the advantages and disadvantages
3. Bean scopes. Name and tell about their life cycle
4. Tools for creating proxies. Differences between Dynamic Proxy and CGLib
5. Spring MVC vs Spring Boot. Differences
6. Distinguishes @Component from @Service and @Repository
7. How to perform a native database query when using Spring Data JPA?
8. How does @Transactional work in Spring? What happens if one service calls a Transactional method from another Transactional method?
9. Name a few popular spring boot starters
10. What is DI and IoC?

Docker:

1. What is the essence of Docker and what problem does it solve?
2. Describe in general terms what should be in the .Dockerfile
3. What is docker-compose and why is it needed?
4. What are layers in images and what problem do they solve?
5. What types of networks does docker-compose allow you to install for services?
6. Case: I launched PostgreSQL in docker, but it is not available on the system via standard port 5432. Where to look and how to diagnose?